Multilinear Algebra

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1 Motivation

Multilinear algebra studies vector spaces and, beyond linear maps, multilinear maps.

We do *not* give spacetime a vector space structure. In the previous lectures a set was equipped with a topology and then with charts. Physical space or spacetime is *not* equipped with a vector space structure: there is no meaning to "five times Paris" or "Paris plus Vienna". Adding places or multiplying them by real numbers is not defined on physical position space.

Why study vector spaces anyway? The *tangent spaces* T_pM of a smooth manifold M (smooth manifolds come in lecture 4, T_pM in lecture 5) do carry a natural vector space structure, even though M itself does not. Once there are vector spaces there are derived notions, namely *tensors*. It is best to study these abstractly first, for two reasons:

1. The construction of T_pM needs an intermediate vector space $C^\infty(M)$, so vector spaces must be understood beforehand.
2. Tensor techniques are defined on any vector space and are easiest to understand in an abstract setting without worrying about a specific space.

2 Vector spaces

Definition 1 (Vector space)**.** A (real) *vector space* is a triple $(V, +, \cdot)$ consisting of a set V and two maps

$$\begin{aligned} + &: V \times V \to V && \text{(addition)},\\ \cdot &: \mathbb{R} \times V \to V && \text{(S-multiplication)}, \end{aligned}$$

satisfying the following eight axioms. (The S-multiplication takes a real number and a vector to a vector; it is not a product of two vectors.)

For all $u, v, w \in V$ and $\lambda, \mu \in \mathbb{R}$:

(1) $v + w = w + v$ (commutativity).

(2) $(u + v) + w = u + (v + w)$ (associativity of $+$).

(3) There exists $\circ \in V$ (neutral element) such that $v + \circ = v$ for all $v \in V$.

(4) For each $v \in V$ there exists $-v \in V$ with $v + (-v) = \circ$.

(5) $(\lambda\mu) \cdot v = \lambda \cdot (\mu \cdot v)$.

(6) $(\lambda+\mu)\cdot v = \lambda\cdot v + \mu\cdot v$.

(7) $\lambda\cdot(v+w) = \lambda\cdot v + \lambda\cdot w$.

(8) $1\cdot v = v$.

Here one must keep track of which operations are which: in (5) the product $\lambda\mu$ is in $\mathbb{R}$, while the dots acting on v are S-multiplication in V; in (6) $\lambda+\mu$ is in $\mathbb{R}$ while the right-hand side uses $+$ in V; and the neutral element $\circ$ of V is not the number $0\in\mathbb{R}$.

Terminology. An element of a vector space is informally called a *vector*. "Is this object a vector?" is not a well-posed question: it is an element of a set, which *may* underlie a vector space, depending on whether addition and S-multiplication are defined. One can only speak of vector spaces. (Similarly for tensors: one defines tensor spaces and informally calls their elements tensors; vectors are a special case. In quantum mechanics a state is naturally an element of a Hilbert space, in fact a ray, not literally "a vector".)

Remark. $\mathbb{R}^3$ is not a vector space by itself: it is a set of triples. It becomes one only after specifying, e.g., componentwise addition and S-multiplication into each entry. This is additional structure, not a "natural" choice just because we are used to it.

2.1 Running example: polynomials

Fix $N\in\mathbb{N}$. Let

$$P = \left\{ p:(-1,1)\to\mathbb{R} \;\middle|\; p(x)=\sum_{n=0}^{N} p_n x^n,\ p_n\in\mathbb{R} \right\}$$

be the set of polynomials of degree at most N on $(-1,1)$. This is only a set, and the question "is $x\mapsto x^2$ a vector?" only makes sense after defining operations. Define

$$(p+_P q)(x) = p(x)+q(x), \qquad (\lambda\cdot p)(x) = \lambda\, p(x),$$

where on the right-hand sides the operations are those of $\mathbb{R}$. Then $(P,+_P,\cdot)$ is a vector space (the axioms are easily checked).

3 Linear maps

Structure is studied through structure-preserving maps (as in topology, where these were continuous maps).

Definition 2 (Linear map)**.** Let $(V,+_V,\cdot_V)$ and $(W,+_W,\cdot_W)$ be vector spaces. A map $f:V\to W$ is *linear* if for all $v,\tilde v\in V$ and $\lambda\in\mathbb{R}$:

(i) $f(v+_V\tilde v) = f(v)+_W f(\tilde v)$,

(ii) $f(\lambda\cdot_V v) = \lambda\cdot_W f(v)$.

We write $f:V\xrightarrow{\sim}W$ for a linear map.

Whether a map is linear can only be asked once both vector space structures are specified.

Example 1 (Differentiation)**.** $\delta: P\xrightarrow{\sim}P$, $\delta(p)=p'$ is linear: the sum rule gives $\delta(p+q)=\delta(p)+\delta(q)$, and $\delta(\lambda p)=\lambda\,\delta(p)$.

Theorem 1. *If $f:V\xrightarrow{\sim}W$ and $g:W\xrightarrow{\sim}U$ are linear, then $g\circ f:V\to U$ is linear.*

For instance $\delta\circ\delta$ (second derivative) is linear. In a basis this fact encodes matrix multiplication.

3.1 The vector space $\mathrm{Hom}(V, W)$

Let V, W be vector spaces. The set

$$\mathrm{Hom}(V, W) = \{f : V \to W \mid f \text{ linear}\}$$

becomes a vector space with

$$(f \mathbin{\hat{+}} g)(v) = f(v) +_W g(v), \qquad (\lambda \mathbin{\hat{\cdot}} f)(v) = \lambda \cdot_W f(v).$$

(The circles mark that these are yet other $+$ and $\cdot$; later they are dropped.) This gives a tower of structures: one can form $\mathrm{Hom}(V, \mathrm{Hom}(V, W))$, etc.

Example 2. $\mathrm{Hom}(P, P)$ is a vector space. It contains δ, $\delta \circ \delta$, etc., and hence also mixed-order operators such as $5\delta + \delta \circ \delta$, which are automatically linear.

4 The dual vector space

Definition 3 (Dual space)**.** For a vector space V, its *dual vector space* is

$$V^* = \mathrm{Hom}(V, \mathbb{R}),$$

the vector space of linear maps $V \to \mathbb{R}$ (where $\mathbb{R}$ is a vector space with its usual addition and multiplication). Its elements are informally called *covectors*.

This is a special but heavily used case of $\mathrm{Hom}(V, W)$: from V and V^* one can construct all the more complicated structures, so they are the elementary building blocks. Whether something is a "vector" or a "covector" only makes sense relative to a chosen vector space V.

Example 3 (Definite integration)**.** Define $I : P \to \mathbb{R}$ by

$$I(p) = \int_0^1 \mathrm{d}x\, p(x).$$

I is linear (the integral splits over sums and pulls out constants), so $I \in P^*$ is a covector.

Remark. Many objects physicists call vectors are really covectors. For instance, the gradient of a function on $\mathbb{R}^n$ is a covector, not a vector. (This is related to the multidimensional chain rule; it will be taken up later.)

5 Tensors

Elements of the tower of structures built from V and V^* are generically called tensors. For finite-dimensional V there is a simple way to define them: as multilinear maps. (The tangent spaces of a d-dimensional manifold will be d-dimensional vector spaces; in general relativity, $d = 4$.)

Definition 4 ((r, s) tensor)**.** Let V be a vector space and $r, s \in \mathbb{N}_0$. An (r, s) *tensor over* V is a multilinear map

$$T : \underbrace{V^* \times \cdots \times V^*}_{r} \times \underbrace{V \times \cdots \times V}_{s} \to \mathbb{R}.$$

Multilinearity (example $r = s = 1$). T is linear in each slot separately:

$$T(\varphi + \psi, v) = T(\varphi, v) + T(\psi, v), \qquad T(\lambda\varphi, v) = \lambda\, T(\varphi, v),$$
$$T(\varphi, v + w) = T(\varphi, v) + T(\varphi, w), \qquad T(\varphi, \lambda v) = \lambda\, T(\varphi, v).$$

Hence $T(\varphi + \psi, v + w) = T(\varphi, v) + T(\varphi, w) + T(\psi, v) + T(\psi, w)$.

5.1 A $(1,1)$ tensor is the same as a linear map $V \to V$

For finite-dimensional V (so $V^{**} = V$ canonically):

- Given $T : V^* \times V \to \mathbb{R}$ multilinear, define $\phi_T : V \to V^{**} \cong V$ by $v \mapsto T(\,\cdot\,, v)$; for fixed v, this is a linear map $V^* \to \mathbb{R}$.
- Conversely, given linear $\phi : V \to V$, define $T(\varphi, v) = \varphi(\phi(v))$.

These constructions are inverse to each other, so both carry the same data. Analogously for general (r, s).

5.2 Examples and special cases

- A covector $\varphi \in V^*$ is a linear map $V \to \mathbb{R}$, hence by definition a $(0,1)$ tensor.
- A vector $v \in V$ can be viewed as an element of V^{**}, i.e. a linear map $V^* \to \mathbb{R}$, hence a $(1,0)$ tensor.
- A linear map $V \to V$ is a $(1,1)$ tensor.
- An inner product on V is a $(0,2)$ tensor. The "inverse" inner product (properly the dual inner product, an inner product on V^*) is a $(2,0)$ tensor.

Example 4. On P, the map $g : P \times P \to \mathbb{R}$,

$$g(p,q) = \int_{-1}^{1} \mathrm{d}x\, p(x)\, q(x)$$

is bilinear and takes no covectors, so it is a $(0,2)$ tensor: an inner product (as familiar from quantum mechanics, where there is also complex conjugation).

6 Bases and dimension

All the above was done without ever choosing a basis or writing a vector as a list of numbers. This is the structure of the theory.

6.1 Choosing a basis

Definition 5 (Basis (Hamel basis))**.** Let V be a vector space. A subset $B \subseteq V$ is a *basis* if for every $v \in V$ there exist a *finite* subset $\{f_1, \dots, f_n\} \subseteq B$ and unique numbers $v^1, \dots, v^n \in \mathbb{R}$ such that

$$v = v^1 f_1 + \cdots + v^n f_n.$$

Remark. This is the basis of linear algebra (Hamel basis). In quantum mechanics one usually uses infinite linear combinations (series), which need a notion of convergence, i.e. a topology on the vector space (a Schauder basis, e.g. the energy eigenfunctions of the harmonic oscillator in a Hilbert space). Those are *not* bases in the sense above; a bare vector space has no topology.

Definition 6 (Dimension)**.** If V has a basis with finitely many elements, say d, then d is called the *dimension* of V, $\dim V = d$ (all bases then have the same number of elements, which is what makes this well defined). Such V is *finite-dimensional*.

For the rest of the course: finite-dimensional spaces are used and bases chosen only when necessary; infinite-dimensional spaces will occasionally be used but never with a chosen basis.

6.2 Components of vectors

Let V be finite-dimensional with a chosen basis $e_1, \dots, e_n$. Then each $v \in V$ is uniquely associated with a tuple $(v^1, \dots, v^n) \in \mathbb{R}^n$, the **components of v with respect to the chosen basis**, where

$$v = v^1 e_1 + \cdots + v^n e_n.$$

A vector has no components by itself, only after choosing a basis. In the space P the vector $x^2 + x^3$ corresponds to many different tuples depending on the basis; the choice of a basis is a lot of information (one among very many choices) that has nothing to do with the vector itself.

Warning. Whatever one writes down using a basis depends on that choice. Concepts defined via a basis (e.g. the determinant or trace of a linear map via a matrix) must be checked to be independent of the choice; better, define them basis-free. Compare with charts: one would only want a *maximal* atlas and make sure everything defined is independent of the chosen chart.

6.3 Dual basis

Given a basis $e_1, \dots, e_n$ of V, one could choose any basis of V^*, but that doubles the arbitrariness. It is more economical to require the **dual basis** $\varepsilon^1, \dots, \varepsilon^n$ of V^*, defined by

$$\varepsilon^a(e_b) = \delta^a{}_b = \begin{cases} 1 & a = b, \\ 0 & a \neq b. \end{cases}$$

This uniquely determines the ε^a from the e_a and makes many formulas simpler.

Example 5 (Polynomials)**.** Let P be the polynomials of degree at most 3 (so $\dim P = 4$) with basis

$$e_a(x) = x^a, \qquad a = 0, 1, 2, 3.$$

The dual basis consists not of polynomials but of derivative operators with evaluation:

$$\varepsilon^a(p) = \frac{1}{a!} \left. \frac{\mathrm{d}^a p}{\mathrm{d}x^a} \right|_{x=0}.$$

Indeed $\varepsilon^a(e_b) = \delta^a{}_b$: the a-th derivative of x^b at 0 is $a!$ if $b = a$ and 0 otherwise (lower powers are killed by the derivative, higher powers vanish at $x = 0$).

7 Components of tensors and the Einstein summation convention

Definition 7 (Tensor components)**.** Let T be an (r, s) tensor over a finite-dimensional vector space V, with a chosen basis $e_1, \dots, e_n$ ($n = \dim V$) and the dual basis $\varepsilon^1, \dots, \varepsilon^n$. The n^{r+s} real numbers

$$T^{i_1 \dots i_r}{}_{j_1 \dots j_s} = T\big(\varepsilon^{i_1}, \dots, \varepsilon^{i_r}, e_{j_1}, \dots, e_{j_s}\big), \qquad i_k, j_l \in \{1, \dots, n\},$$

are the *components of T with respect to the chosen basis.*

The components, together with the basis, determine the whole tensor.

Reconstruction (example $(1,1)$). Expand $\varphi = \varphi_i \varepsilon^i$ and $v = v^j e_j$. By multilinearity,

$$T(\varphi, v) = \sum_{i=1}^{n} \sum_{j=1}^{n} \varphi_i \, v^j \, T(\varepsilon^i, e_j) = \sum_{i,j} \varphi_i \, v^j \, T^i{}_j.$$

Einstein summation convention. Label basis vectors with lower indices (e_j), dual basis covectors with upper indices (ε^i), components of vectors with upper indices (v^j), components of covectors with lower indices (φ_i), and tensor components with upper indices for the V^*-slots and lower indices for the V-slots. Then in expressions every index that appears once up and once down is automatically summed over. The summation signs above become invisible:

$$T(\varphi, v) = \varphi_i \, v^j \, T^i{}_j.$$

Hiding sums is safe because one always deals with multilinear maps, for which a sum can be pulled in or out.

Summary

- Spacetime itself is not a vector space; tangent spaces will be.
- A vector space is a set with addition and S-multiplication satisfying eight axioms; "vector" just means an element of a vector space.
- Linear maps preserve the structure; $\mathrm{Hom}(V, W)$ is itself a vector space; $V^* = \mathrm{Hom}(V, \mathbb{R})$ is the dual space (elements: covectors).
- An (r, s) tensor is a multilinear map $(V^*)^r \times V^s \to \mathbb{R}$. Vectors are $(1, 0)$, covectors $(0, 1)$, linear maps $V \to V$ are $(1, 1)$, inner products are $(0, 2)$.
- Bases and components are a choice with no objective meaning; define things basis-free where possible. The dual basis links the choice on V^* to that on V.
- Tensor components are $T^{i_1 \cdots}{}_{j_1 \cdots} = T(\varepsilon^{i_1}, \ldots, e_{j_1}, \ldots)$, with the Einstein summation convention.